ECS7032P Assignment 1 – Individual Reflection

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Group report: Decoupled Information-Set MCTS with a Card-Value Model for Sushi Go!

My role. I carried most of the research, an equal share of the work next to Hakan's coordination and Halit's code. I studied how Sushi Go! is modelled in TAG and what makes it hard for plain MCTS: hidden hands, simultaneous card choices and several players. I then read the literature on each problem: determinisation and information-set MCTS for hidden information, decoupled UCT and its alternatives for simultaneous moves, max-n backups for several players, and simulation balancing for rollout policies. I summarised what each method would mean for our agent, and these summaries were the basis of our main technical decisions: one tree node per turn in which every player chooses from its own statistics, hidden cards re-sampled in every iteration instead of a few fixed samples, and an evaluation based on the lead over the best opponent, since only first place wins.

My teammates. Hakan led the group, divided the workload and led our research and decisions. Halit handled the code side: the agent in TAG, the experiments and the competition files.

Reflection. The research showed me that the right method depends on the game. Decoupled selection over information sets improved our agent by 8.2 percentage points over its sequential, determinised predecessor, while TAG's own agent of this kind, without our card model, was not detectably better than our previous version. I also learned to read every result together with its error bars. Next time I would also test the alternatives to decoupled UCT, regret matching and EXP3, which our report leaves as future work.

Use of generative AI: as our report states, we used Claude (Anthropic) for the code and the drafting; this reflection was also drafted with Claude from our own account of the work.

Word count: 288 (without the heading)